How Ask Captain Jim Scores Sailing Conditions

A plain-English walkthrough of the scoring engine, for explaining the app to others.

1. The Big Picture

Every time you open the app (or hit "Refresh Live Data"), Captain Jim does the same four things:

- Pulls live wind, gust, rain, and storm data from the National Weather Service for your marina's exact location.
- Pulls live tide predictions from NOAA for the nearest official tide station.
- Runs every hour of today (and the next 6 days) through a scoring formula, using the sailing preferences you've set.
- Finds the best stretch of hours to sail, and boils the whole day down to one number from 0–10 and a plain label: Excellent, Good, Marginal, or No-Go.

Nothing is guessed or estimated from historical averages — every score is calculated fresh from that day's actual forecast and tide predictions.

2. Where the Data Comes From

Data	Source	What it provides
Wind & weather	National Weather Service (api.weather.gov)	Hour-by-hour wind speed, gusts, rain chance, and storm warnings, specific to your marina's coordinates
Tides	NOAA Tides & Currents	Hour-by-hour predicted water height at the nearest official tide station

The "nearest tide station" is chosen by straight-line distance to a fixed list of NOAA stations for the region. That's a reasonable approximation almost everywhere, but it doesn't account for which side of a bay or inlet you're actually on — so occasionally the closest station by miles isn't the most locally accurate one.

3. Scoring a Single Hour

Every hour of the day gets its own score from 0 to 10. Think of it as starting at a perfect 10 and losing points for anything that makes conditions less pleasant or less safe — unless something crosses a hard safety line, in which case the hour is scored a flat zero regardless of anything else.

Hard stops (automatic zero)

An hour is scored 0 — no partial credit — if any of these are true:

- Wind is below 4 knots (not enough to sail) or above 26 knots (too rough, regardless of preference)
- Gusts exceed the maximum gust you've told the app you're comfortable with
- Chance of rain is 70% or higher
- The forecast text mentions thunderstorms, severe weather, or tropical systems

Point deductions (if it clears the hard stops)

Starting from 10, points come off for:

Condition	Effect on score	Why it's weighted this way
Wind below your preferred minimum	−0.5 per knot short	Too little wind means slow, frustrating sailing — annoying but not dangerous, so it's a moderate penalty
Wind above your preferred maximum	−0.6 per knot over	Too much wind is penalized slightly harder than too little, since overpowered conditions escalate toward being unsafe, not just less fun
Gusts more than 7 knots above sustained wind	up to −2.5	A big gap between steady wind and gusts means unpredictable, punchy conditions — harder to sail comfortably even if the average wind looks fine
Gusts within 3 knots of your personal maximum	−1.5	A warning penalty for cutting it close to your own comfort ceiling, before it becomes an automatic zero
Chance of rain	−1 point per 25% chance (capped at −3)	Rain matters, but capped so a modest rain chance doesn't overwhelm an otherwise great wind day
Tide moving the direction you prefer (incoming/outgoing)	+0.5 bonus	A small reward, not a requirement — tide direction is a nice-to-have preference, not a safety factor

Tide moving the opposite direction from your preference	−0.3	A mild penalty, intentionally smaller than the bonus, so tide preference nudges the score without dominating it

The final hourly score is capped between 0 and 10.

4. Finding the Best Window to Sail

The app doesn't just average the whole day — it looks for the best *contiguous* block of hours matching how long you want to sail (4 hours by default), only considering daylight hours between 7am and 6pm.

For each possible window of that length, it blends the hours together as:

Window Score = (0.7 × average of the hours) + (0.3 × the worst hour in that stretch)

The 30% weight on the single worst hour is deliberate: it means one rough hour drags the whole window down, even if the rest of the window looks great on average. This favors *consistently* good conditions over a window that's mostly great with one bad patch — which matters more on the water than it would on paper, since you can't just skip the bad hour.

The window with the highest blended score becomes your "Best Departure" / "Best Return" time.

5. Turning the Number into a Label

The winning window's score becomes the day's headline number, and gets translated into a plain-English label:

Score	Label
9.0 – 10.0	🟢 Excellent
7.0 – 8.9	✦ Good
5.0 – 6.9	🟡 Marginal
Below 5.0	🔴 No-Go

This same process runs independently for each of the next 7 days, so the 7-day view is really seven separate best-window calculations, not a rough weekly guess.

6. What You Control

The scoring formula itself is fixed, but it's personalized through your Sailing Preferences:

Preference	Default	What it changes
Preferred wind range	8–18 knots	Where the −0.5 / −0.6 per-knot penalties start applying

Maximum gust you'll accept	22 knots	Where the hard zero and the −1.5 "cutting it close" penalty kick in
Sail duration	4 hours	The length of the window the app searches for
Tide preference	Any (no preference)	Whether incoming or outgoing tide earns the small bonus/penalty
Marina / location	Tampa – Hooker Point	Which weather grid point and tide station feed the whole calculation

In short: two sailors looking at the exact same weather can see different scores and different "best windows," because the formula is the same but it's being run through each person's own comfort settings.

7. Honest Limitations

- The 4–26 knot hard safety bounds are fixed for everyone and aren't adjustable, even though comfort with wind varies a lot by boat and experience.
- The nearest tide station is picked by straight-line distance, not by which station best represents the water you're actually sailing on.
- The forecast is only as good as NWS's hourly model — the app doesn't second-guess or blend in other forecast sources for its primary source (a backup source is used only if NWS is unavailable).